Calculation of Optimized Pulse Patterns for Electric Drives with an End-To-End Differentiable Simulation Framework

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Simulation framework

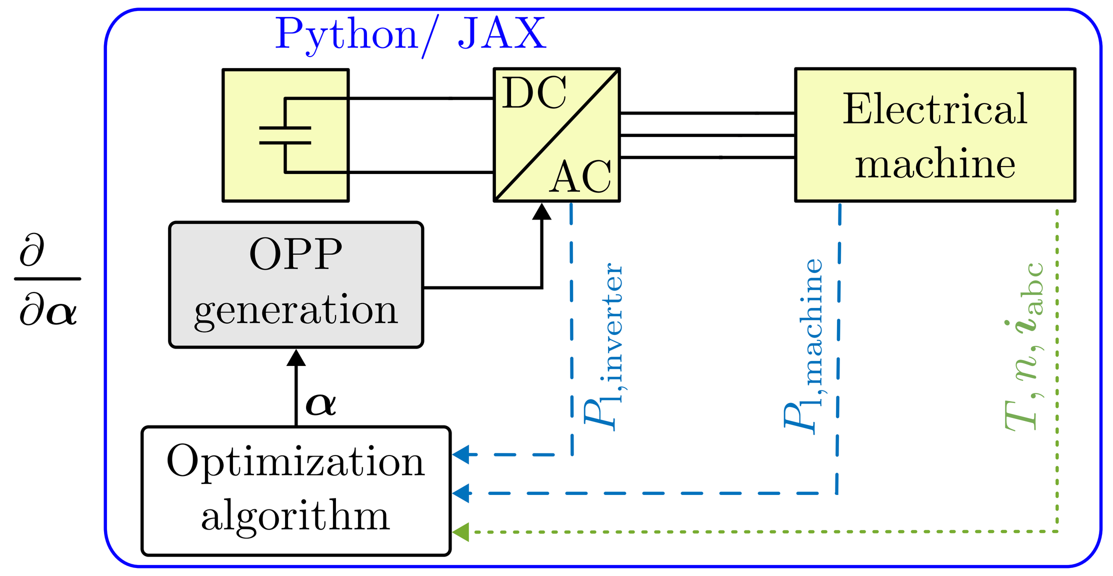

Figure 1. End-to-end differentiable simulation framework.

Contribution

- End-to-end differentiable simulation to calculate OPPs
- Utilization of the nonlinear machine model
- Considering the inverter and machine losses in the cost function

Differential inductances

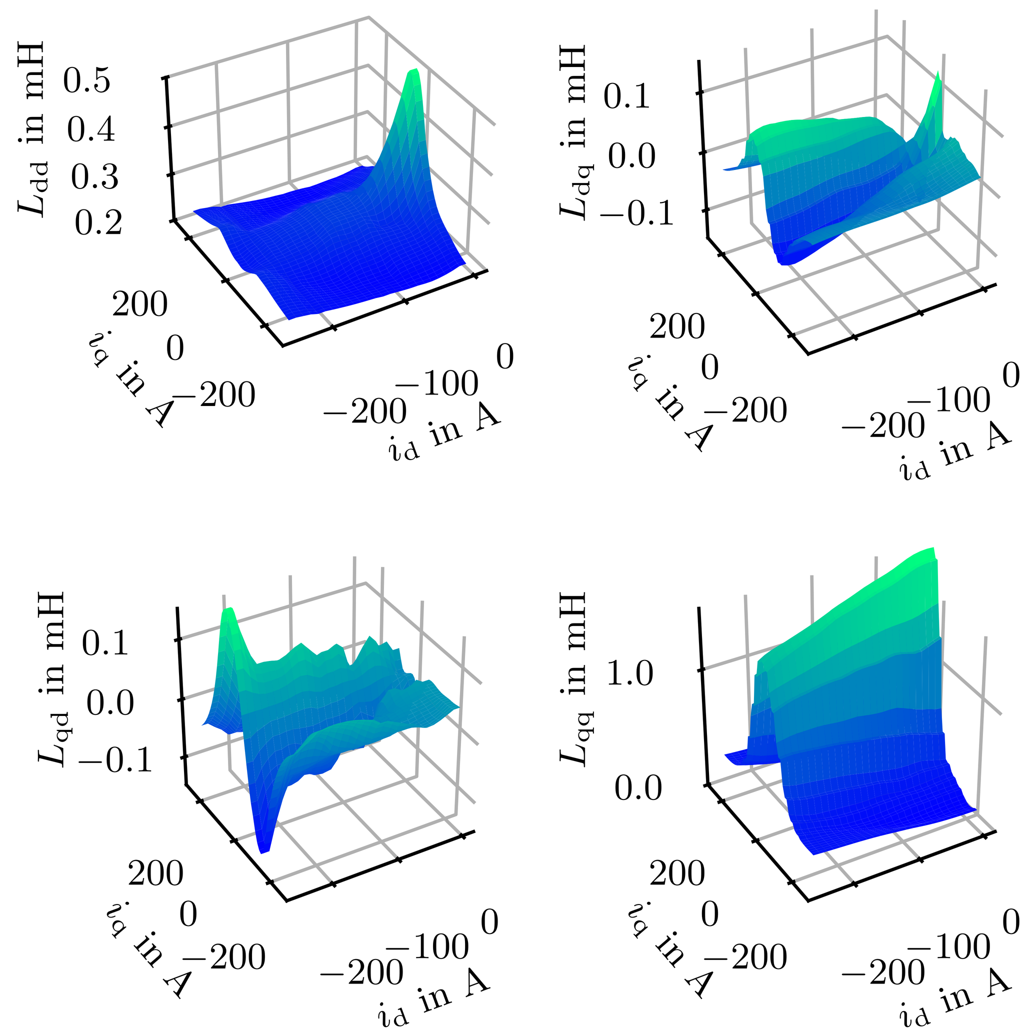

Figure 2. Differential inductances of the utilized PMSM.

Modulation index

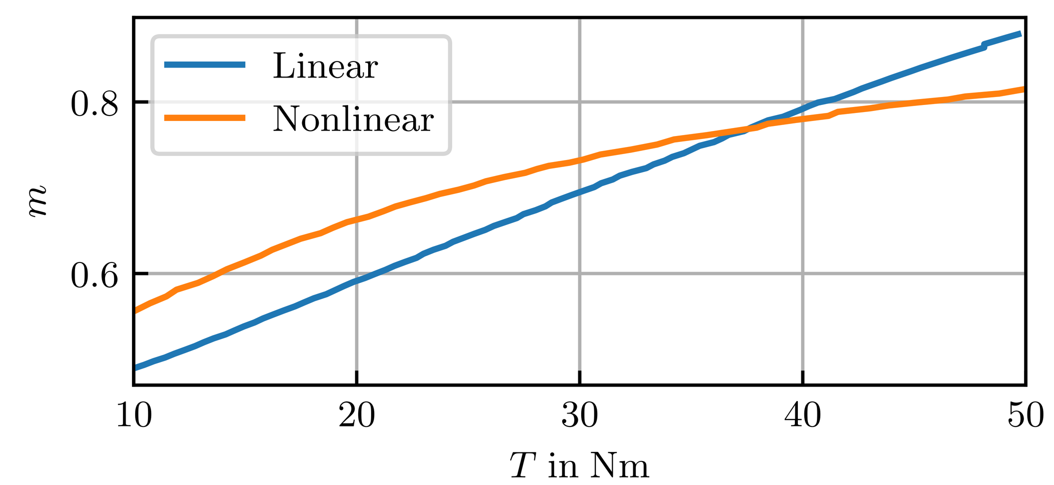

Figure 3. Comparison of the modulation index for the linear and nonlinear PMSM model.

Cost function

$$
\begin{aligned}
\min_{\boldsymbol{\alpha}} \; & J = k_{\mathrm{l,inv}} P_{\mathrm{l,inverter}} + k_{\mathrm{l,mach}} P_{\mathrm{l,machine}} \\
\text{s.t.} \; & \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}), \quad T = T^* \\
& 0 \le \alpha_1 \le \ldots \le \alpha_M \le \frac{\pi}{2}
\end{aligned}
$$

Pulse pattern

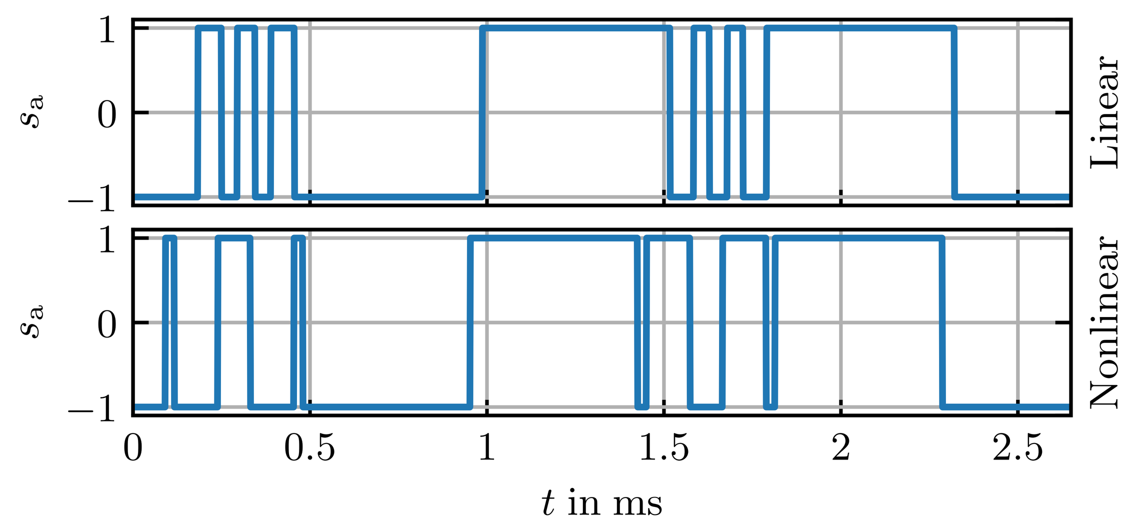

Figure 4. Corresponding pulse patterns to the current trajectories in Fig.5.

Stator current

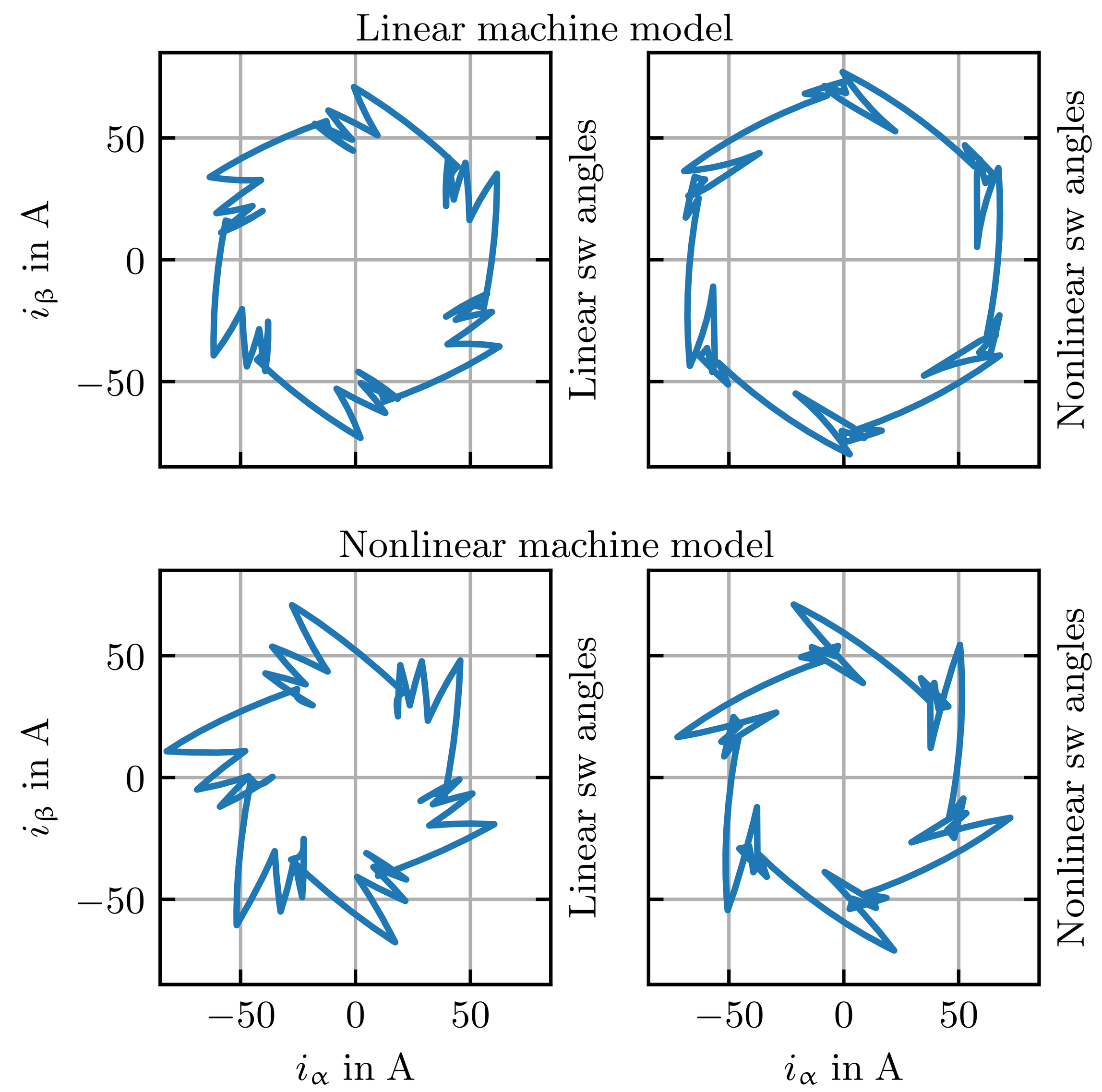

Figure 5. Current trajectories for one operating point to highlight the difference between the linear and nonlinear PMSM model.

Table 1. Specification of the operating points.

$T^* \in [10, 40]$ Nm	
$n^* = 7750$ min^{-1}	
Number of initial start angles	500
Iterations optimizer per evaluation	100

Table 2. Computational effort of the optimization for one operating point (OPP3). The simulations are performed on the CPU of a workstation (AMD Ryzen 9 5950X @3.4 GHz) with 16 cores.

OPP3	Machine model	
	Linear	Nonlinear
Trials	90	70
Average time per operating point	5.27 min	28.07 min
95th percentile time	5.29 min	29.10 min

Switching angles

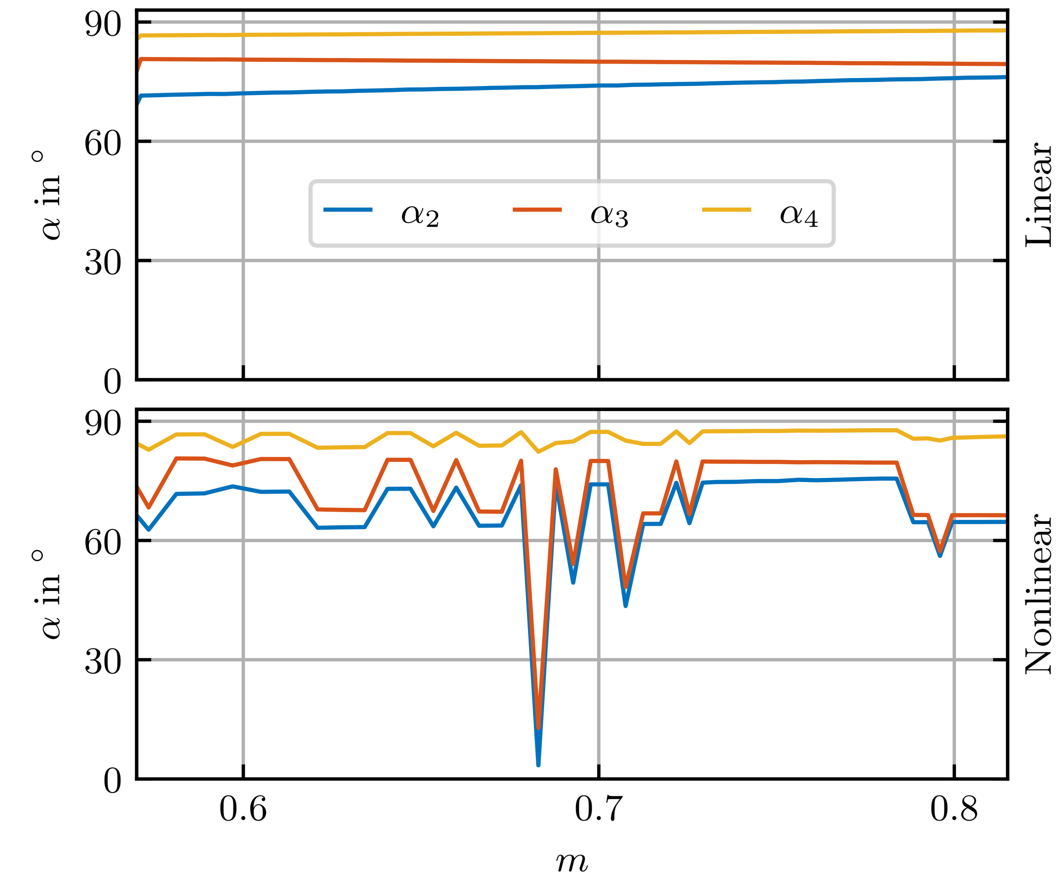

Figure 6. Comparison of the switching angles between a linear and nonlinear PMSM model for a fix rotational speed of $n = 7500\ \mathrm{min}^{-1}$ and $T^* \in [10, 50]$ Nm.

Multi-objective optimization

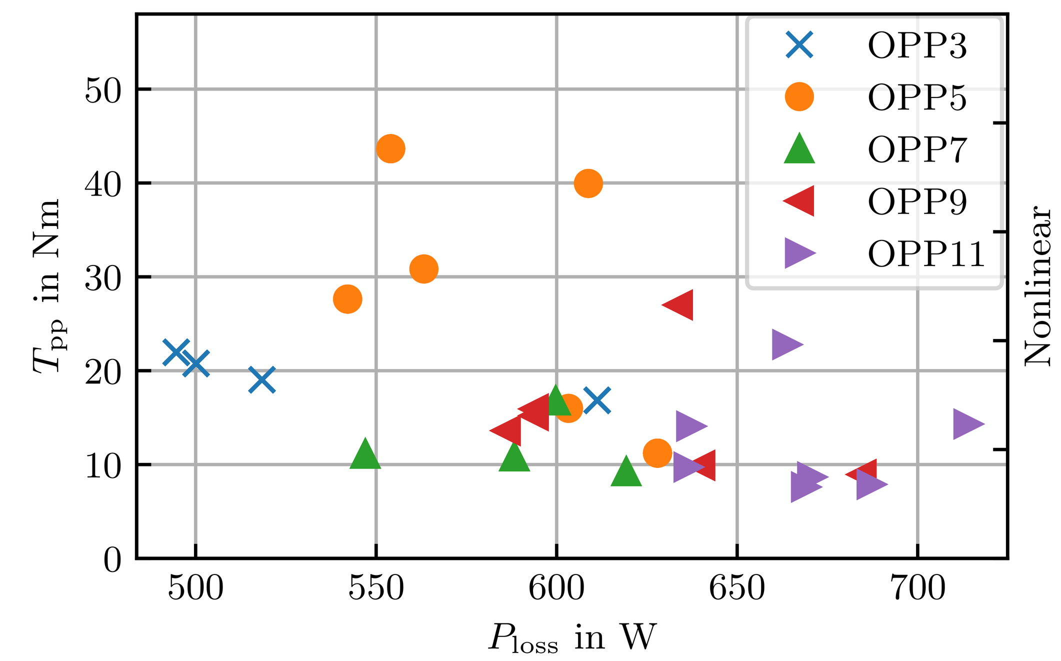

Figure 7. Pareto plane for one specific operating point with $n = 7500\ \mathrm{min}^{-1}$ and $T^* = 40$ Nm. Much more simulations have been carried out as depicted in the figure. However, they result in the same minima.

Conclusion

- Highlight the difference between the linear and nonlinear machine model by comparison of the stator current trajectories
- Reducing the electric drive losses of approx. 8 % and torque ripple of 39 %

Contact

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